REPORT 03 / GLASS BUILD / 12 NOVEMBER 2026

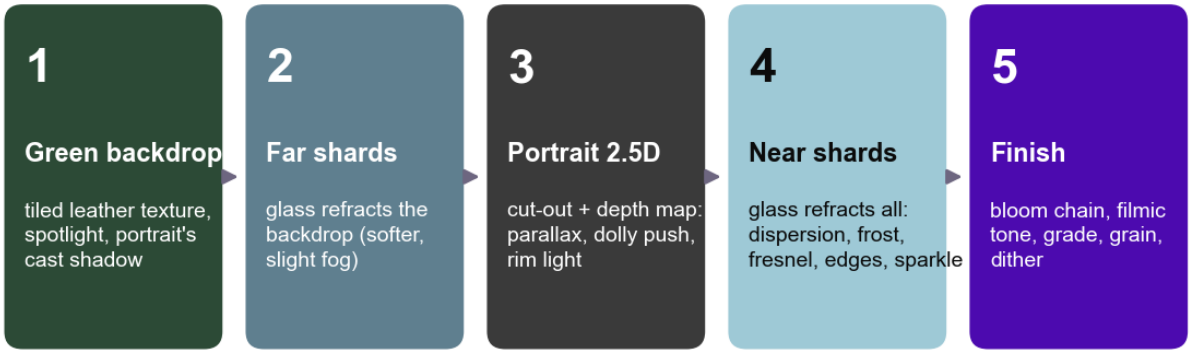
GLASS, GREEN, DEPTH.

Cards became refractive glass shards that drift sideways; the backdrop is a green textured wall; your portrait is cut out and given 2.5D depth.

DEMO	scroll-hero-frame/index.html (card version kept as index-cards.html)
ENGINE	scroll-hero-glass.js (WebGL2, no dependencies)
TESTED	1600 x 832 desktop, 375 x 812 phone, in the app's browser
STATUS	Working. Shard/texture choices use your own recreated assets (licence note in section 6).

1. What changed

You asked for	Built as
Fewer cards	11 pieces instead of 34: 6 near + 5 far (phones: 4 + 3). data-front / data-back change it.
Cards turned into glass with textures, from the THE_DAY_ZERO_MAIN shard models	Real shard meshes from your own theday_glass_shards.glb (same mesh names as glass_shards.glb). Each is screen-space glass: refraction of everything behind it, red/blue dispersion, your frost texture as a refraction map and visible crack highlights, the frost normal map as surface detail, fresnel sheen, bright edges and a sparkle highlight.
Sideways motion	Shards travel left to right with scroll at their own speeds, tumble between two random poses, wobble slowly, and keep drifting sideways when you stop.
Green textured background	A seamless green pebbled-leather tile (own-made to match your reference tile), lit by a soft spotlight behind you, with a vignette.
Portrait: no floating, spatial depth, background removed, features unchanged	The portrait stands grounded at the bottom edge. Its background is removed (only transparency added; pixels untouched). A depth map drives a 2.5D parallax and dolly push as you scroll or move the mouse; it casts a soft shadow on the wall and gets a thin rim light.



Layering:

far shards sit behind you (the portrait is drawn over them), near shards in front; the text layer and CSS shade stay on top.

Shard meshes: your own theday_glass_shards.glb (Shard_01-07 near, BG_Shard_01-09 far), read by a built-in GLB parser.

Glass recipe: THE DAY GlassShardPass (normal + UV mask, then refraction composite), plus the frost normal map.

Figure 1.1 What the canvas draws every frame.

2. The result

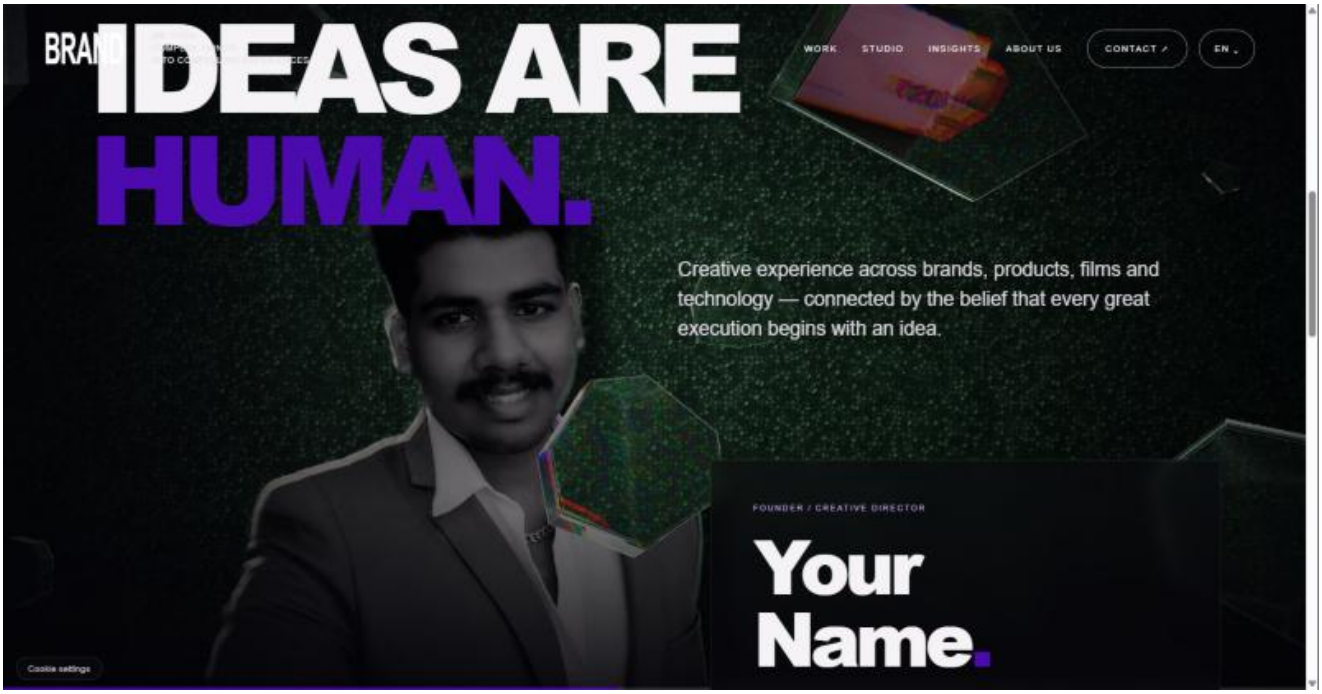


Figure 2.1 Section entering: headline over the green wall, glass drifting past, a rupee image inside a near shard (right).



Figure 2.2 Mid-scroll: profile card, portrait with rim light, shards crossing above the head.

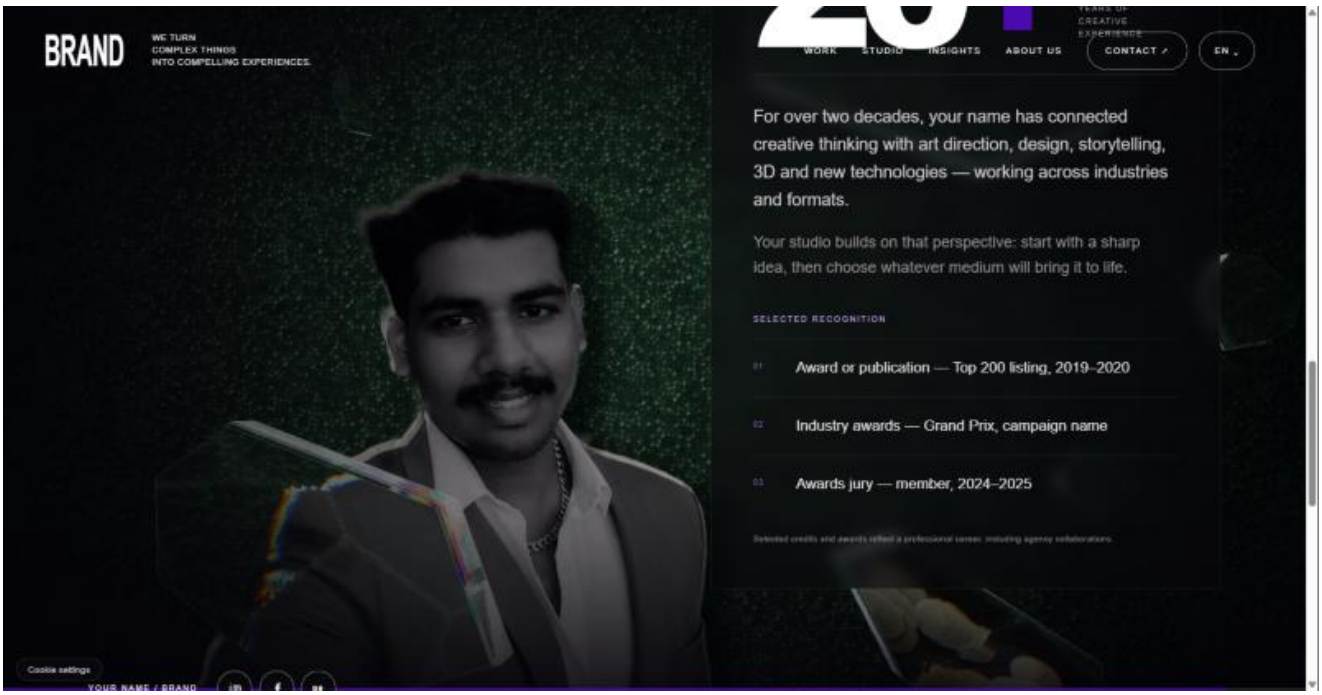


Figure 2.3 Late scroll: awards list; a near shard refracts the jacket (lower left); a far shard sits behind the portrait (right).



Figure 2.4 Glass close-up: refraction, colour fringing at the edges, frost highlights, an image held inside a shard.

3. Your portrait

Background removed offline with OpenCV: the photo's background is out of focus, so a sharpness map seeds GrabCut, the hair (dark on a dark sky) is seeded separately, and a guided filter turns the hard mask into a soft edge that follows the hair. **Only the alpha channel is new; every visible pixel of the face is the original.** A depth map (white = near) is built from the silhouette with a dome on the face.



Figure 3.1 Original | cut-out (checkerboard = transparent) | depth map used for the 2.5D effect.

Second pass (after your feedback that the trim looked off)

The first cut kept a grey haze of blurred sky around the hair, clipped the left ear and left a pale fringe on the jacket. The second pass seeds both ears, tightens the soft edge so the haze drops out, and decontaminates edge colour (estimates the background behind each edge pixel and removes it). Interior pixels are still the original photo.



Figure 3.2 Before (left) and after (right) on the green wall: hair halo, ears, shoulder edge.

How the depth effect works

- **Parallax:** as the camera pans (scroll moves it slowly left to right, the mouse adds a little), near parts of the face shift more than the edges, so the head reads as round, not flat. Strength is capped at about 1% of the image width so features never warp.
- **Dolly push:** over the scroll the portrait grows slightly, near parts more than far parts.
- **Grounding:** it is anchored to the bottom edge and casts a soft shadow onto the wall up and to the right, so it sits in the space instead of floating.
- **Light:** a thin warm rim on the edges facing the light, a faint green bounce on the far side, slight depth shading. No retouching.



Figure 3.3 Phone: the portrait centred in the video panel, glass crossing the chest.

4. Glass and motion

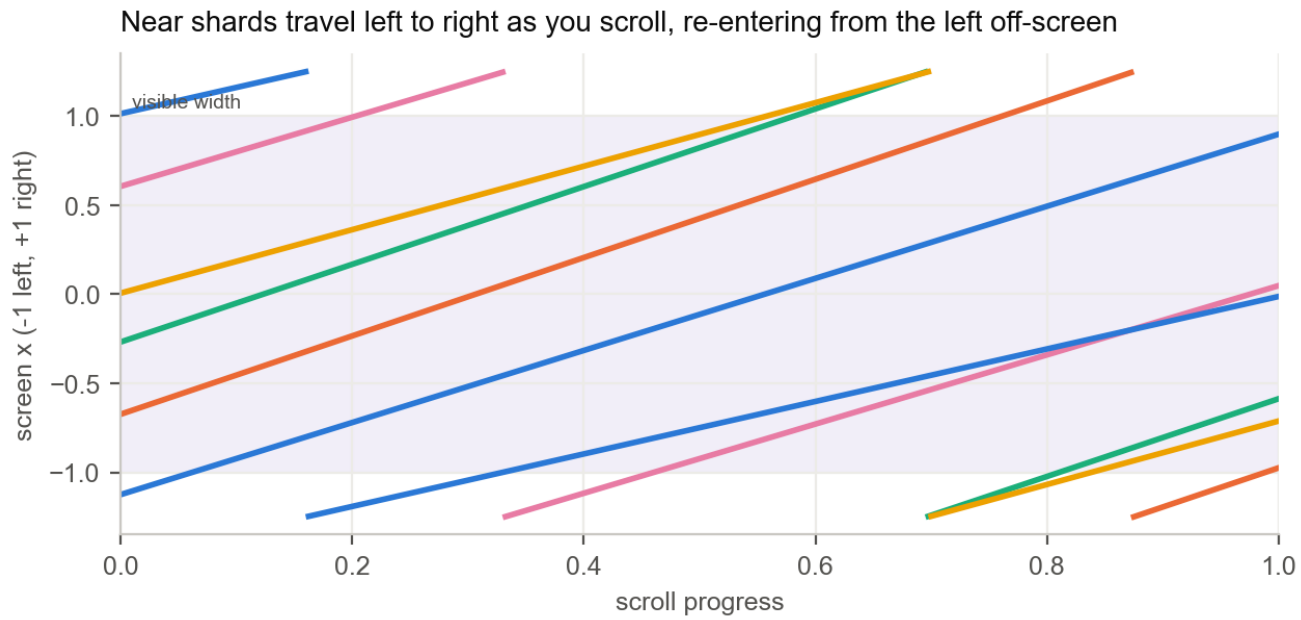


Figure 4.1 Each line is one near shard. They cross the screen left to right as you scroll, then re-enter from off-screen left.

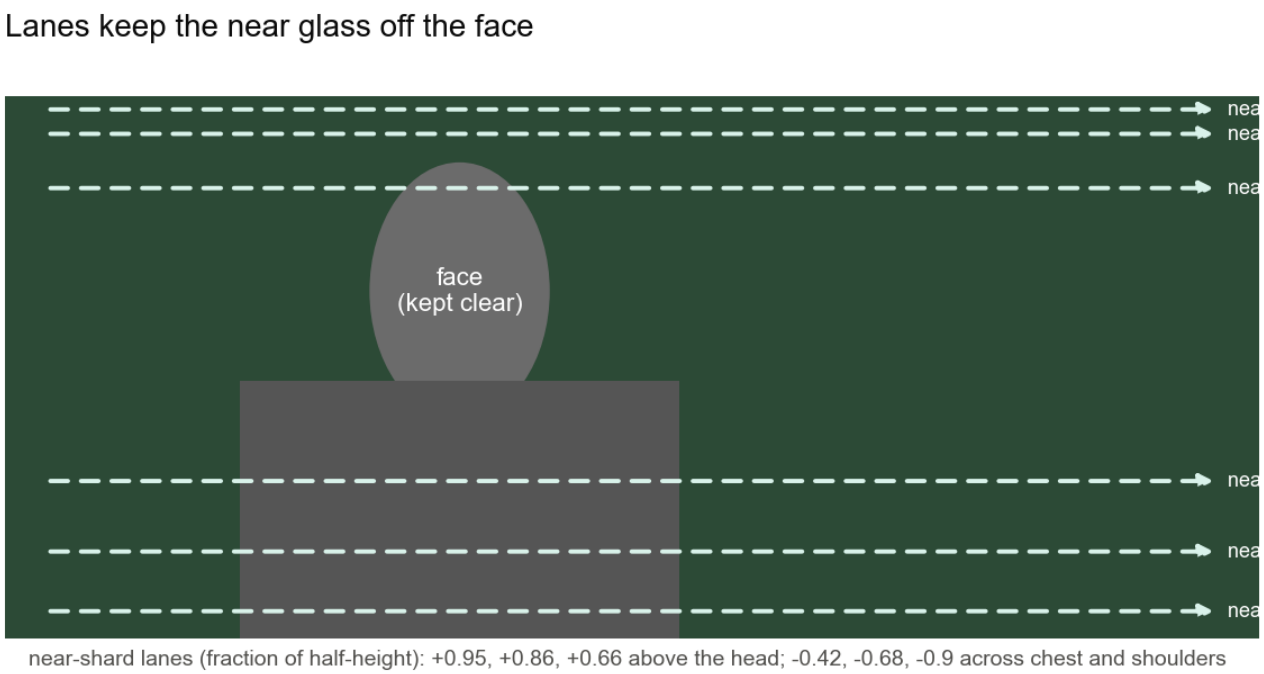


Figure 4.2 Near shards use lanes above the head and across the chest, so glass never sits over your face.

Glass parameter	Near	Far	From
Refraction	0.14	0.08	THE DAY 0.25, reduced for larger shards
Dispersion (R/B split)	0.12	0.08	THE DAY 0.15
Fresnel power	5	6	THE DAY 10, softened
Edge highlight	0.22	0.10	THE DAY 0.15
Frost cracks / sparkle	0.22 / 0.7	0.12 / 0.25	new
Normal-map detail	0.6	0.5	frost-normal.png

5. Textures

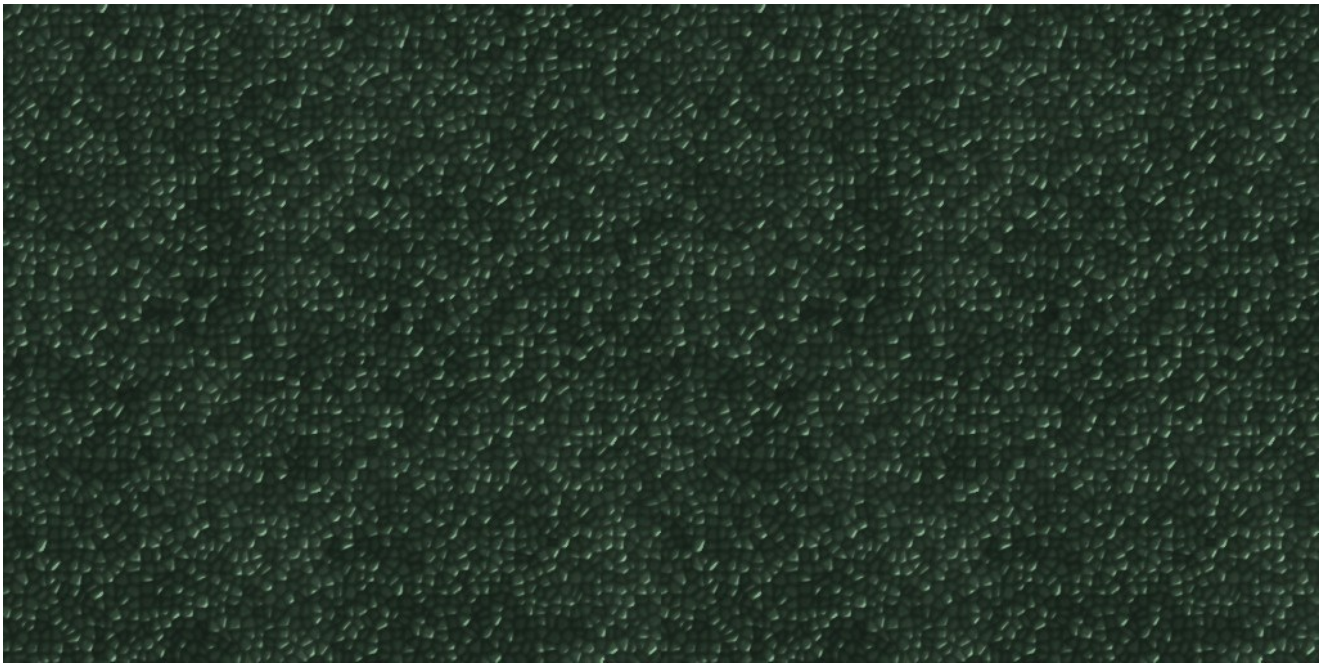


Figure 5.1 Own-made seamless green pebbled-leather tile (two tiles side by side: no seam).

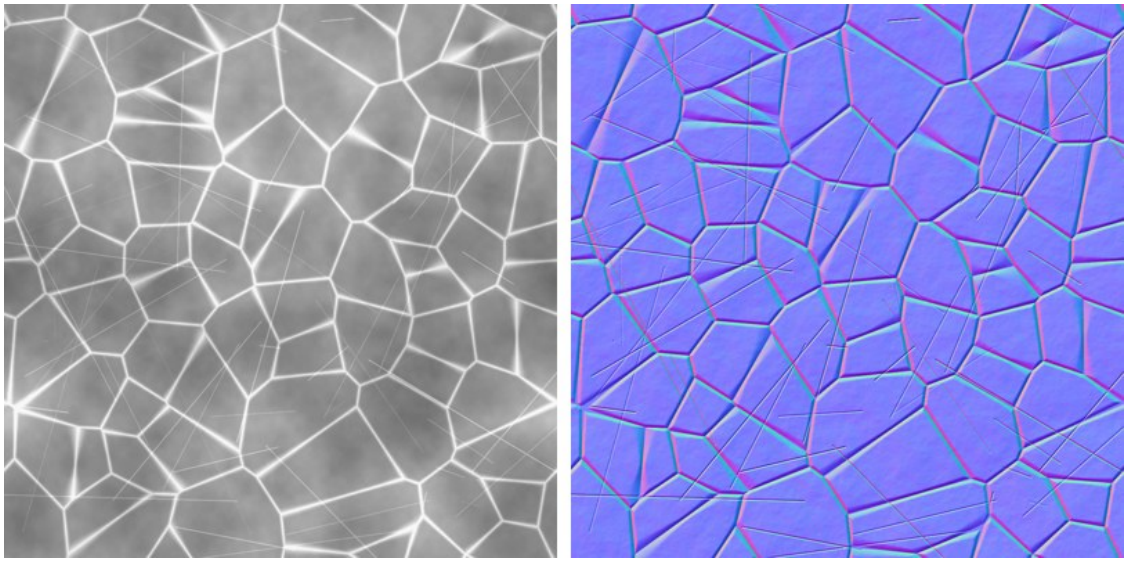


Figure 5.2 Your own frost texture (refraction map + crack highlights) and frost normal map (surface detail).

6. Assets and licence note

Your `THE_DAY_ZERO_MAIN/ASSETS_LICENSES.md` marks the why.zero.university mirror (`reference_originals/`, `reference_textures/`) as **private-only**. Because this frame is meant for other projects, it ships with your own recreations:

Used in the frame	Source	Note
assets/glass/shards.glb	own_recreated/models/theday_glass_shards.glb	drop-in for glass_shards.glb (same mesh names)
assets/glass/frost.jpg / frost-normal.png	own_recreated/textures/theday_frost*.png	your own

Used in the frame	Source	Note
assets/bg/green-leather.jpg	generated (tools/make_green.py)	matches stage2_background2 in colour and grain
assets/subject/portrait*.png	Downloads/Cinematic Noir Portrait at Dusk-1.png	your photo; cut out with tools/cutout_portrait.py
stage2_glass-shatter.glb	reference original	not used (private-only, no own version yet)

For a private build you can point the frame at the originals: `data-shards` and `data-background` take any path.

7. How to use it

```
<div class="shf-scene"
     data-portrait="assets/subject/portrait.png"
     data-depth-map="assets/subject/portrait-depth.png"
     data-background="assets/bg/green-leather.jpg"
     data-shards="assets/glass/shards.glb"
     data-frost="assets/glass/frost.jpg"
     data-frost-normal="assets/glass/frost-normal.png"
     data-front="6" data-back="5" data-subject-x="0.36">
  <img src="assets/cards/rupee-01.jpg" alt="">   <!-- optional: images held inside near shards -->
</div>
<script src="scroll-hero.js" defer></script>
<script src="scroll-hero-glass.js" defer></script>
```

New portrait:

```
python tools/cutout_portrait.py "path/to/photo.png" assets/subject/portrait
```

Option	Default	Effect
data-front / data-back	6 / 5	number of near / far shards
data-subject-x	0.36	portrait centre across the frame (desktop)
data-portrait-height	1.04	portrait height relative to the frame
data-depth	1	strength of the 2.5D effect (0 = flat)
data-speed	1	sideways travel speed
data-tile	360	background tile size in px
data-bloom / data-grain	0.7 / 0.03	finish

8. Verification

Check	Result
Loads with no console errors	pass: 16 shard meshes parsed, 11 shards placed, 3 images, all textures
Desktop 1600 x 832, scroll 0.25 to 0.97	pass (section 2); loop runs at the display rate
Phone 375 x 812	pass: centred portrait panel, 7 shards
Face kept clear of near glass	pass after lane change (figure 4.2)
Fallback without WebGL2	code path only: green tile + still portrait via CSS; not tested on such a device
Physical phones	not tested

Not done / your call

- Hair-edge quality: the second offline cut is clean, but fine flyaway strands are simplified. A dedicated matting model (rembg) would keep them; it needs a one-time download of about 170 MB, so it is your call.
- stage2_glass-shatter.glb (the pre-fractured animated pane) is not used: it is a private-only reference file and there is no own version yet.
- The card version is untouched in index-cards.html + scroll-hero-cards.js.